

for a \$5 billion investment on a fab will probably take place in five years, but an ATMP will deliver return on investment probably in just three years,” explains Krishna.

A peek into history shows that the first ATMP units were not started as a necessity. In fact, there were no major players involved in the ATMP business during the initial years. None of the big players got into ATMP or OSAT units. “This was something that big players didn’t want to get into during the early days. The view was that this work was better done by someone else as there was no superior technology involved. Moreover, it was more labour-intensive,” says Raja.

Innovation in packaging

Many people, especially in India, view packaging as a lower stack. It’s usually the design engineers that are seen on the top of the totem. The manufacturing engineers are usually placed on the middle or the lower end of the cycle, and the test engineers are seen as the last on the stack. However, Raja is of the view that the real innovation will take place in packaging and testing part. “Tremendous amount of innovation will happen in the test and packaging domain. For example, we are trying to put antennas inside the package. We are doing silicon photonics,” says Raja.

This innovation in packaging is one more reason that explains India’s interest in the domain because the country and its engineers are already well known for their design capabilities. “India is best suited as a country for ATMPs and OSATs because of the design capabilities it has. What started as a dumb manufacturing once has now evolved to engineering-intensive manufacturing, and that makes India the best suited country,” says Raja.

The fact that almost every big semiconductor company of the world has an R&D office situated in India cements what Raja says. The likes of Broadcom, NXP, Samsung, Applied Materials, Micron, and many others have offices located in the country. Intel, as a matter of fact, has its second biggest design centre in Bengaluru.

According to the Department of Electronics and Information Technology (DeitY), nearly 2,000 chips are being designed every year in India,

and more than 20,000 engineers are working on various aspects of chip design and verification. More stats by the ministry reveal that the foreign direct investment (FDI) in electronic manufacturing was 1,230 billion rupees (approx. US\$18.34 billion) in 2016. It was only 110 billion rupees (approx. US\$1.64 billion) in 2014.

But what is the primary reason behind India’s interest in semiconductors? Krishna answers, “It is because the country is facing this much trouble in product manufacturing due to shortage of semiconductors.”

This is a time when even individuals not associated with electronics are asking questions around the shortage of semiconductors, and how soon can India solve that problem. A recent report, mentioned by Krishna, has placed electronics at higher value than gold, and oil is the only item that supercedes electronics as per that report.

Adding fuel to the fire are the geopolitical conditions that have arisen during the last two or three years. These geopolitical conditions involving China and the United States have made the policy makers worldwide, and not just in India, to look for alternatives to their needs. Almost every country is searching for a second option, and this second option has to be a country with high knowledge and understanding of the subject matter, whether it is semiconductors or medical equipment.

“The design ecosystem that India has built looks like a perfect launchpad for what needs to happen next. Fab is at the other end of it but there are things in between that need to happen, and that is why we are talking about ATMPs and OSATs today,” explains Krishna.

There are global design centres located in India that have not limited its designers to the design aspect but have also handed over the architecture of chips to Indian engineers. The practice earlier was entirely different as architects were usually from outside India. The biggest factor, as Krishna notes, is companies like Tata saying that we will have to cut down our production by billions of dollars owing to the semiconductor shortage in the world.

Krishna jokingly adds, “It is all about timing in cricket and semicon-

ductors. You can’t hit boundaries if you do not time the ball!”

Can ATMPs/OSATs make a difference

A lot of professionals, as well as experts sitting in media houses, are of the view that setting up OSATs or ATMPs is of no use unless there are fabs in a country. However, many of these experts often forget that most of the top 20 semiconductor companies of the world are fabless. “Setting up a local OSAT or an ATMP unit can definitely help in fostering a robust ecosystem of semiconductors in India,” says Venkata.

The question that pops up here is that if setting up an OSAT or ATMP can really help then why India has been missing the bus time and again? There have been instances when big names have shown interest in setting up such units but nothing has gone beyond paper and site surveys.

“India has done quite well in terms of mobile manufacturing during the last six years, and with Apple setting up two factories in India, the focus on the electronics manufacturing sector is getting more attention in the country,” says Venkata. He adds, “It is a clear proof that the country now has started believing in its potential. However, the issue is that neither the components nor the semiconductors used in manufacturing these electronics are made in India.”

Additionally, the government of India had recently invited expression of interest from companies interested in setting up fabs in India. However, as per Venkata, that looks like a distant dream. ATMPs and OSATs can make a difference if the industry instead of going after latest technologies goes after use cases, ones that will be the most adopted in India.

“OSAT can definitely be the right start for building a semiconductor ecosystem in India but we cannot give up on bringing or establishing fabs in India,” says Venkata. He adds, “Maybe the five billion dollar fabs can wait but there are other technologies that we should focus on bringing to India.”

The one fact that designers keep in mind while working on new semiconductors is that they are not designing these for just one market but for applications all over the world.